

DIBAKAR ROY

 Portfolio |  dibakar0804 |  roy056 |  Google Scholar

Panchagarh, Rangpur, Bangladesh

RESEARCH INTERESTS

Neural engineering, brain-computer interfaces, neural signal decoding, EEG signal processing, biomedical machine learning, and adaptive intelligent systems.

EDUCATION

• Rajshahi University of Engineering & Technology (RUET)

BSc. in Electrical & Computer Engineering

◦ CGPA: 3.39 / 4.00 | (Last 60 Credits: 3.99 / 4.00)

March 2022 - July, 2026

Rajshahi-6204, Bangladesh

• Notre Dame College

Higher-Secondary School Certificate (HSC)

◦ GPA: 5.00 / 5.00

2018 - 2020

Dhaka-1000, Bangladesh

RESEARCH EXPERIENCE

• Thesis Research – Rapid Cross-Session Calibration for Intracortical Speech Decoding

- Developed a parameter-efficient cross-session decoding framework on the T12 and T15 attempted-speech datasets using chronological evaluation.
- Implemented a collapse-resistant CTC decoder with input projectors, a residual adapter, and a two-layer BiGRU; calibration updates only lightweight components while freezing the encoder and CTC head.
- Improved all six held-out T12 sessions using only 80 labeled calibration trials per session, with statistically significant paired session-level improvement.

• Baseline-Aware EEG Workload Classification with Reduced Electrodes

- Developed a baseline-aware, leakage-safe framework for subject-independent workload classification on EEGMAT and STEW using nested LOSO validation.
- Evaluated reduced and shared montages, handcrafted and Riemannian features, and cross-dataset transfer; accepted as a conference book chapter.
- Achieved balanced accuracies of 0.8199 on EEGMAT and 0.8519 on STEW under strict nested LOSO evaluation.

PUBLICATIONS

• Hardware-Aware RIS-Assisted ISAC with Quantized Feedback: A Robust DRL Framework.

Published in *Proceedings of the IEEE International Conference on Computer and Information Technology (ICCIT 2025)*. 

• Learning-Augmented RIS-Aided Hybrid Beamforming for Secure and Low-Latency Transmission.

Published in *Proceedings of the IEEE International Conference on Electrical, Information and Communication Technology (EICT 2025)*. 

• Fast Label-Free Cross-Session Calibration for Intracortical Speech Neuroprostheses.

Published in *Proceedings of the IEEE International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN 2026)*. 

• Baseline-Aware Practical Framework for EEG Workload Classification with Reduced Electrodes Across Multiple Datasets.

Accepted and presented at the International Conference on Electrical, Computer and Communication Technologies (ECCT 2026); camera-ready manuscript submitted in Taylor & Francis book-chapter format for publication.

ACADEMIC PROJECTS

- **Event Management Website (Software Project II)**

2024, RUET

- Built a centralized departmental event management system with admin/user dashboards, secure authentication, and real-time updates.
- Implemented using Next.js and the MERN stack with role-based access control and email verification.



- **Woman Safety Device (Electronic Shop Practice)**

2023, RUET

- Designed an ESP32-based safety device integrated with a custom Android SOS app via Bluetooth.
- Implemented live GPS location sharing, incident logging for analysis, and a built-in taser for primary self-defense.

- **Nutrition Website (Software Project I)**

2023, RUET

- Developed a health-focused web platform for BMI, BMR, and daily calorie estimation based on user activity.
- Implemented OTP-based email verification for secure user registration and login.



EXPERIENCE

- **Industrial Trainee**

BJIT Ltd.

March 2025

Dhaka, Bangladesh

- Completed industrial training at BJIT Academy with hands-on exposure to SDLC and Agile/Scrum practices.
- Developed Python-based applications including an OOP-driven Library Management System and a Task Manager with JSON persistence.
- Gained practical experience with Git/GitHub, software quality assurance concepts, and team-based software development.

SKILLS

- **Programming:** Python, MATLAB, C++, JavaScript
- **Signal Processing:** MNE-Python, EEG preprocessing, spectral analysis
- **Research Tools:** Git, GitHub, L^AT_EX, Overleaf, Jupyter, VS Code
- **Engineering & Design Tools:** Proteus, EasyEDA, AutoCAD, AutoCAD Electrical
- **Web Development:** MERN Stack, Next.js, HTML, CSS
- **Documentation & Productivity Tools:** L^AT_EX, Microsoft Office

ACHIEVEMENTS

- Merit-Based Scholarship during BSc. from the Directorate of Technical Education, Bangladesh.
- General Scholarship (SSC) and Talentpool Scholarship (JSC & PSC).
- Second Runner-Up (Junior Category) in the Regional Bangladesh Mathematical Olympiad (BDMO) and participant in the National Round.
- Champion, Inter-Department Volleyball Tournament, RUET.



LEADERSHIP, SERVICE & ACTIVITIES

- **Class Representative (2nd Year)**

Department of Electrical and Computer Engineering

- Represented the class in academic and administrative matters and coordinated communication between students and faculty.



- **Volunteer**

trECento, American Center Pop-Up & Inter-Department Project Showcasing

- Supported event organization, coordination, and logistics for departmental programs, the U.S. Embassy-supported American Center Pop-Up, and university project showcases.

- **Sports**

Department of Electrical and Computer Engineering

- Represented the department in inter-department football, volleyball, fresher's football, and cricket tournaments.